

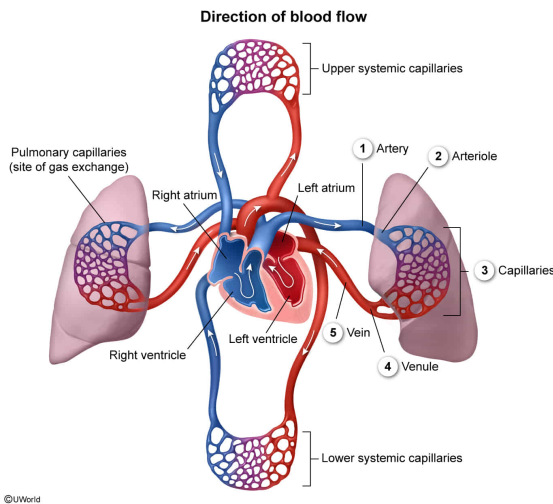
Which series shows the order in which blood passes through the pulmonary circulation?

- ☐ A. Vein → venule → capillary → arteriole → artery [13%]
- ☐ B. Venule → vein → capillary → artery → arteriole [2%]
- ☐ C. Arteriole → artery → capillary → vein → venule [3%]
- ✓ ☒ D. Artery → arteriole → capillary → venule → vein [79%]

Correct

78% answered correctly

Explanation:



The **blood vessels** of the circulatory system are divided into **two circuits**: the systemic circuit and the pulmonary circuit. The systemic circuit distributes nutrient-rich blood that is high in O_2 and low in CO_2 to all body tissues. The pulmonary circuit moves deoxygenated blood (drained from the systemic circulation) to the lungs to become reoxygenated.

When deoxygenated blood departs the right ventricle of the **heart**, it takes the following path:

1. Blood first enters **arteries** (ie, pulmonary arteries), or vessels that carry blood away from the heart.
2. The arteries, through which the blood flows, become progressively smaller in diameter until they branch into **arterioles**.
3. Arterioles deliver blood to the **capillaries** (smallest blood vessels) of the lungs. **Gas exchange** occurs across these capillaries: deoxygenated blood in the capillaries expels CO_2 to the alveolar spaces (air sacs), which will be exhaled from the body. The blood in turn receives O_2 from the alveolar spaces (ie, becomes reoxygenated).
4. From the lung capillaries, freshly oxygenated blood enters the smallest veins (vessels that carry blood toward the heart), known as **venules**.
5. Venules become progressively larger **veins** (ie, pulmonary veins), which are low-pressure vessels that will ultimately deliver blood to the left atrium of the heart. From the left atrium, blood will enter the systemic circuit and carry the oxygenated blood to

systemic tissues for [systemic gas exchange](#).

Therefore, the correct order in which blood passes through the pulmonary circulatory system is artery → arteriole → capillary → venule → vein **(Choices A, B, and C)**.

Educational objective:

In the pulmonary circulation, blood passes from the heart through progressively smaller arteries and arterioles until it arrives at the lung capillaries for gas exchange with the inspired air. Afterward, blood is carried back to the heart via venules and larger veins to be pumped through the systemic circulation to the tissues.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration